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Q1

1 OLS estimate: $\sum \mu_i^2$ has the minimum

$\mu_i = y_i - \beta_0$

so $\sum (y_i - \beta_0)^2$ takes the minimum

$$\frac{d}{d\beta_0} = -2\sum (y_i - \beta_0)$$

when $\frac{d}{d\beta_0} = 0$ It is the minimum.

so $\sum (y_i - \beta_0) = 0$

$$\Rightarrow \frac{\sum (y_i - \beta_0)}{n} = 0$$

$\beta_0 = \bar{y}$ so the OLS estimator of β_0 is $\bar{y}$

2. $\mu_i = y_i - \beta_1 x_i$

$$\sum \mu_i^2 = \sum (y_i - \beta_1 x_i)^2$$

$$\frac{d}{d\beta_1} = -2\sum x_i (y_i - \beta_1 x_i)$$

when $\frac{d}{d\beta_1} = 0$ $-2\sum x_i (y_i - \beta_1 x_i) = 0$

so $\sum x_i y_i - \beta_1 \sum x_i^2 = 0$

so $\beta_1 = \frac{\sum x_i y_i}{\sum x_i^2}$

2.

1. $\beta_1 > 0$ Because I think more education can improve people's capacity, which gives them more chance to find a job with higher wage.

2. $\hat{\beta}_1 = 17910.03$ $se = 193.9109$

$\hat{a} = 5740 5.06$ $se = 2004.897$

$R^2 = 0.1225$

Yes, agree

3. $\hat{\beta}_1 = 17910.03$ means that if the years of education improves 1 year, monthly wages will improve 17910.03

$a = \hat{\beta}_0 = 5740 5.06$ means the monthly wage when years of education is 0.

4. Pass

5. $\widehat{Wage} = \alpha + \beta_1 \times 10$

$= 5405.06 + 1791.03 \times 10$

$= 23505.36$

6. $R^2 = 0.1225$

It means that 12.25% is explained by their education.

Reasons:

1. Economics: Rural families may have poorer economics situation, can't afford to educatio

2. Transportation: In rural area, people may live far away from the school, the poor transportation may make people hard to attend school.

70 for

log-linear:

$\log(wage)_i = 11.41764 + 0.073147\ education_i$

the parameter 0.073147 means that the wage increases 7.3147% for every addition year of education

② for

linear-log

wage i = - 493 08.24

1285(4.6 log(education)

the parameter 12851f .6 means that the wage increases by 1285.146 for 1% change in years of education

③ for log-log

log(wage):

= 10.95083 + 0.54084492 log(education)

the parameter 0.540842 means that the wage increases by 0.540842% for every 1% increase in sales

8. SLR.4 is $E(u|education) = 0$

it means that the value of residual contains no information of education.

Use STATA to

reg (education residuals)

we find that the

coefficient between education

and residual is -2.02×10^{-14}

close to 0, and $p = 1$

means u and education

have almost no linear

relation. So $E(u|\text{education}) = 0$

The assumption holds.

Q3

- μ_i represents the differences between predicted yields value and real yields value.

The land quality may make different plots have different μ_i

2. $E[\mu_i | X_i] \neq 0$ in this case, since the amount of fertilizers may effect soil's quality, which means μ_i is correlated with X_i

3.

(a) No, it may be biased because it could reflect differences in unobserved characteristics rather than

the true effect of fertilizer.

(b) The yields will take both effects of soil quality and fertilizers, which will result in a biased estimate

(c) Each plots' soil quality and their amounts of fertilizers received.

Use MLR:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 S_i + \mu_i$$

S_i = Soil quality measure for the i-th plot

Y_i = Yield of ith plot